# Week #8: [Muscles of the Arm]

*(Non-hyperlinked = Dr. Caitlin Cassidy Module)*

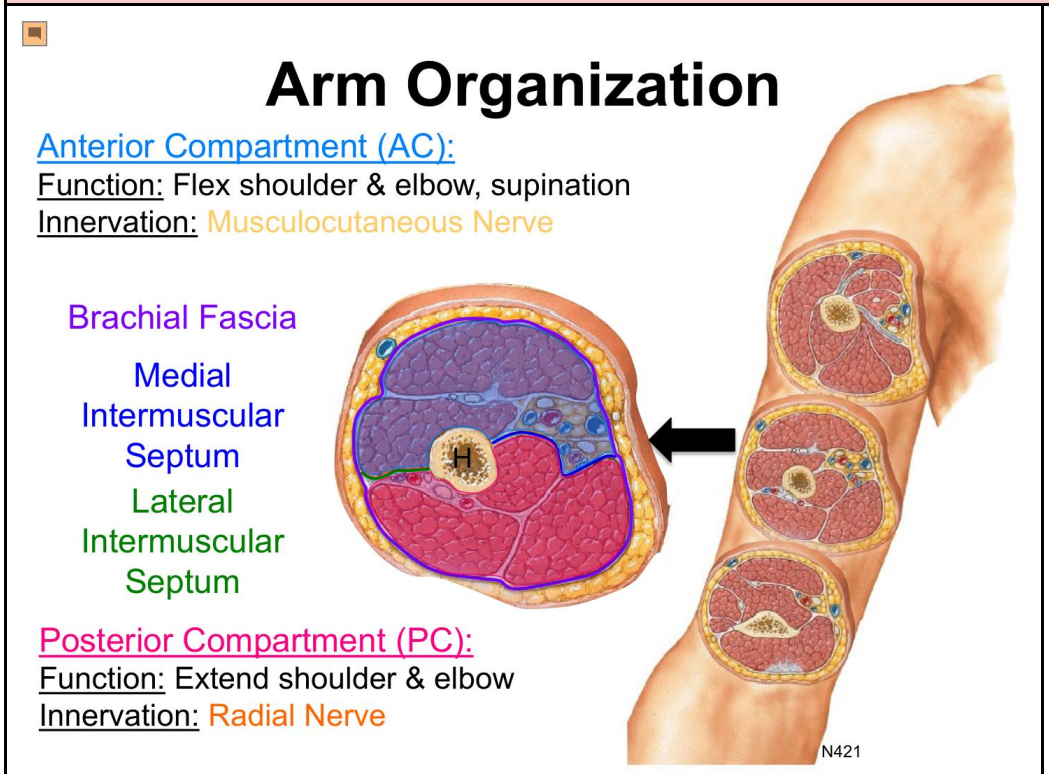
• Introduction to Module

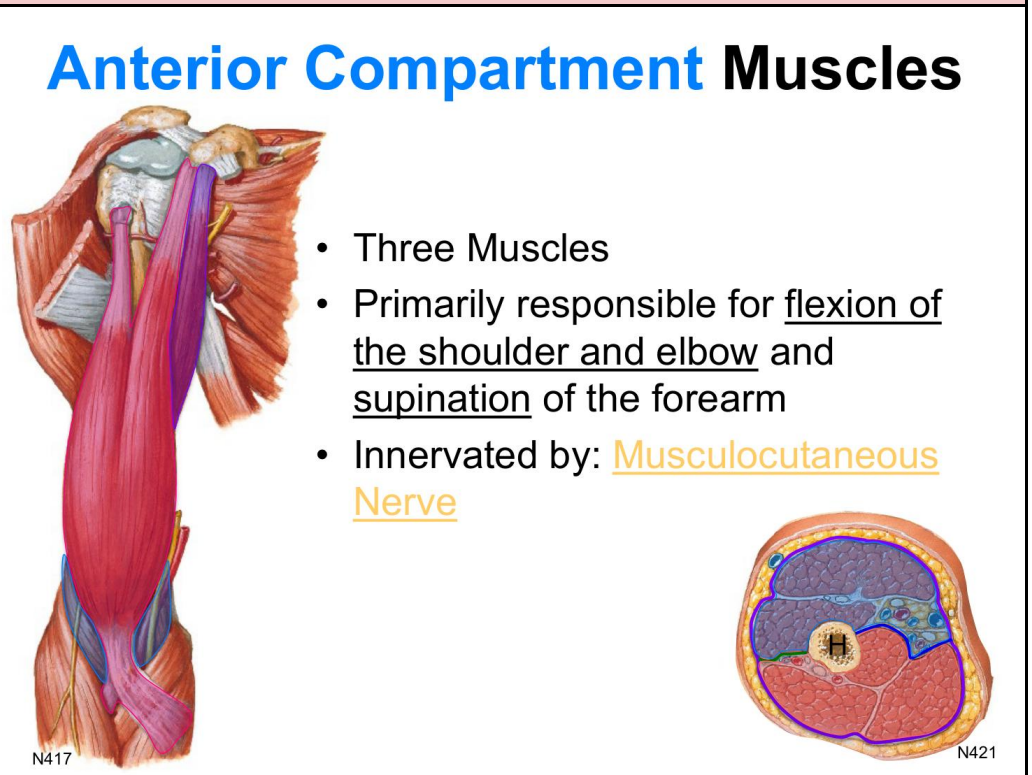
**Muscles of the Proximal Upper Extremity**

- **Arm Musculature (23 min)**
- **Approach to Proximal Biceps Brachii Tendinopathy (10 min)**

**Arm Musculature**

- Muscles of the Arm




**Arm Organization:** The arm extends from the shoulder to the elbow. Deep to the skin and subcutaneous tissue is the deep fascia of the arm, known as **brachial fascia.** Two intermuscular septa (***medial and lateral intermuscular septa***) extend from the brachial fascia to the humerus to divide the arm into two compartments:

- Anterior compartment
- Posterior compartment

These compartments divide the arm into functional and innervation compartments

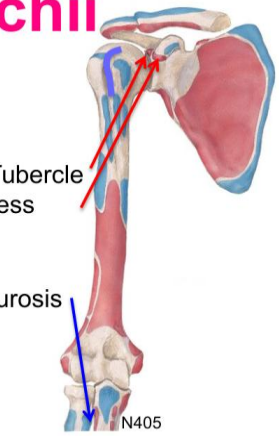
# AC Muscles: Biceps Brachii

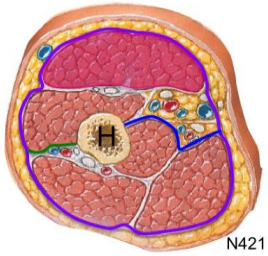



Origin:
Long head: Supraglenoid Tubercle
Short head: Coracoid Process

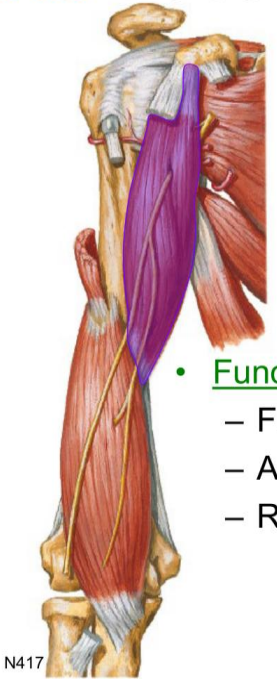
Insertion:
Radial Tuberosity & Aponeurosis

- Function:
  - Strong supinator
  - Flexes forearm
  - Weak shoulder flexor
  - Resists dislocation



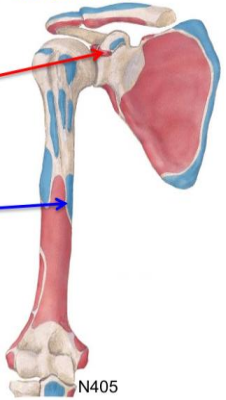

# AC Muscles: Coracobrachialis

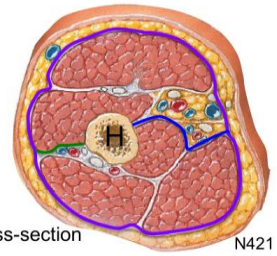



Origin:
Coracoid Process

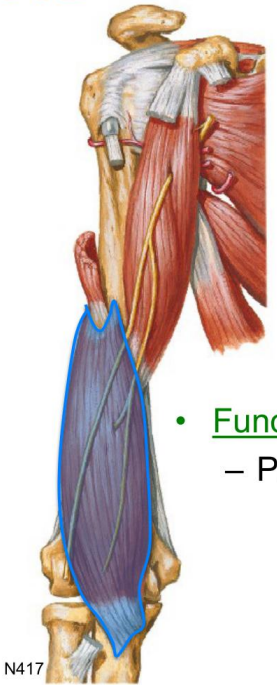
Insertion:
Mid ⅓ of Medial Humerus

- Function:
  - Flexes arm
  - Adducts arm
  - Resists dislocation



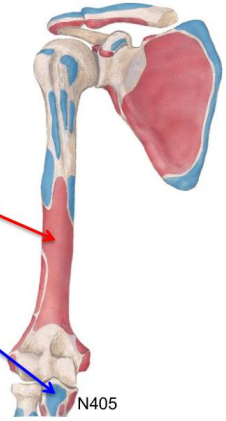

# AC Muscles: Brachialis

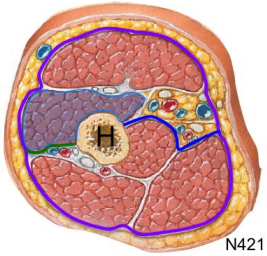



Origin:
Distal ½ of Anterior Humerus

Insertion:
Coronoid Process of Ulna

- Function:
  - Primary flexor of the forearm

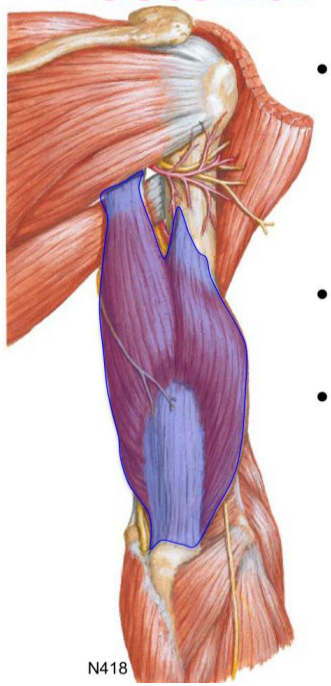


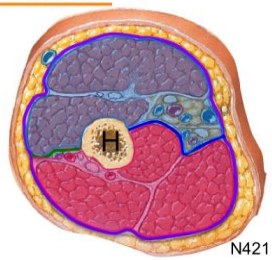
**Anterior Compartment Muscles:** function to ***flex the shoulder and elbow*** as well as ***supinate*** the forearm and they are all innervated by ***musculocutaneous nerve.***

| Muscle | Origin | Insertion | Innervation | Function |
|---|---|---|---|---|
| Biceps Brachii | LH: Supraglenoid tubercle of scapula<br>SH: Coracoid process | Radial tuberosity and forearm via bicipital aponeurosis | Musculocutaneous Nerve | Supinate & flex forearm; flex shoulder; resist dislocation |
| Coracobrachialis | Coracoid process | Mid 1/3 medial humerus | | Flex and adduct shoulder; resist dislocation |
| Brachialis | Distal ½ anterior humerus | Coronoid process of ulna | | Primary elbow flexor |

## Posterior Compartment Muscle

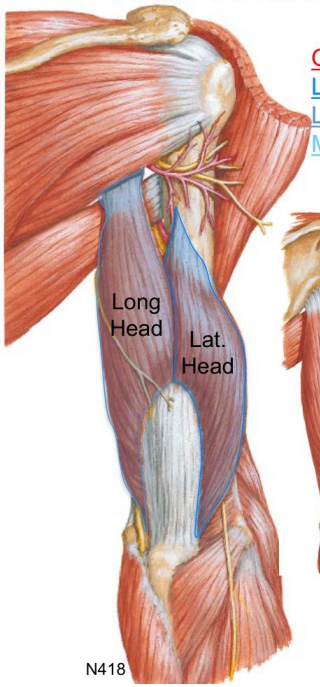


- One Muscle: Triceps
  - Three Heads:
    - Long Head
    - Lateral Head
    - Medial Head
- Primarily responsible for extension of the shoulder and elbow
- Innervated by: Radial Nerve

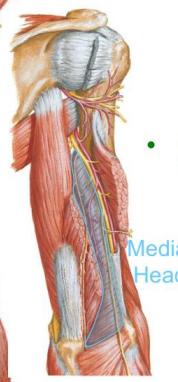


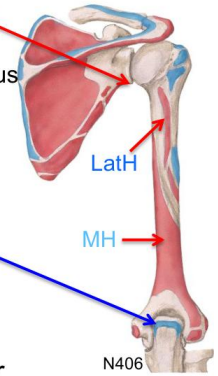

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## PC Muscle: Triceps Brachii




Origin:
Lateral Head: Proximal ⅓ of Posterior Humerus
Long Head: Infraglenoid Tubercle
Medial Head: Distal ⅓ of Posterior Humerus

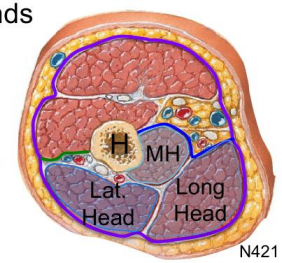



Insertion:
Olecranon Process

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- Function:
  - Primary Elbow Extensor
  - Long Head: extends shoulder joint & resists shoulder dislocation




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**Posterior Compartment Muscles:** function to e***xtend the shoulder and elbow*** and they are all innervated by ***radial nerve.***

# Knowledge Check

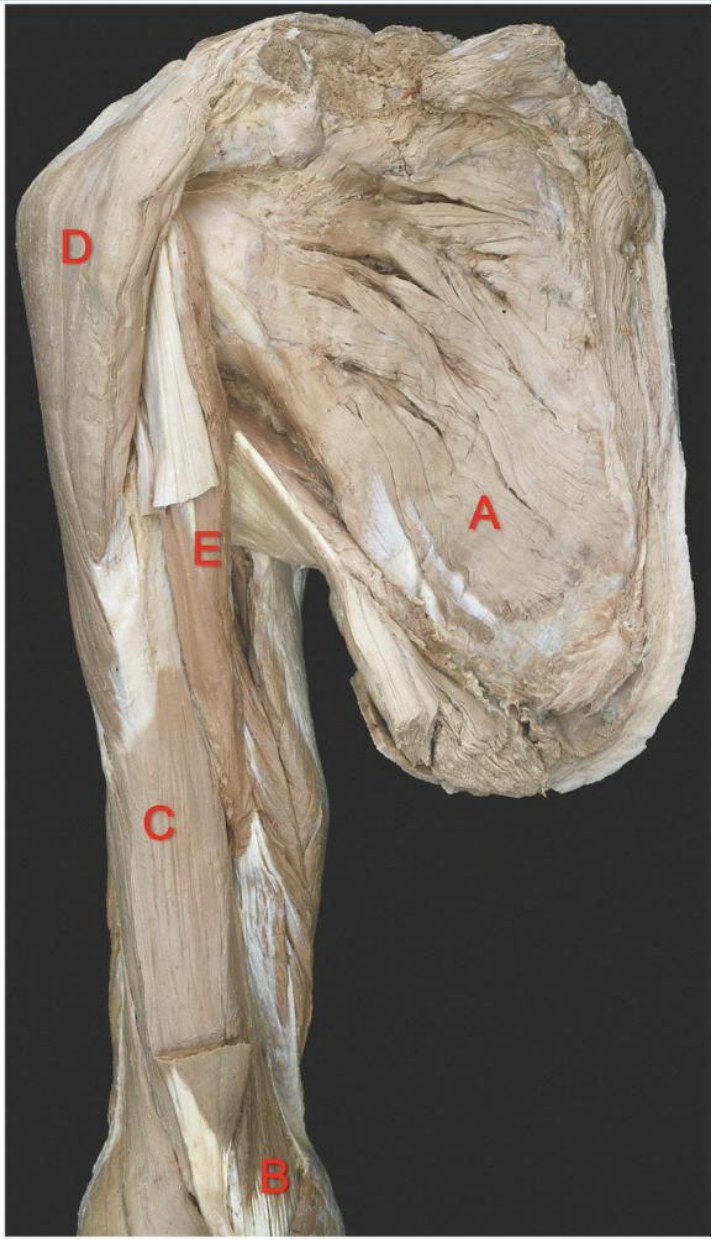
Biceps Brachii C

Brachialis B

Coracobrachialis E

Deltoid D

Subscapularis A



# Muscles of the Proximal Upper Extremity

Approach to Proximal Biceps Brachii Tendinopathy

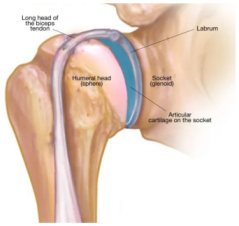
## Proximal Biceps Tendinopathy

- Common cause of shoulder pain
- "Tendinopathy" not "tendinitis"
  - Spectrum of potential abnormalities ranging from overuse/degeneration ("tendinosis"), to acute inflammation, to partial or complete tearing
- Can affect anyone!
  - Younger population – athletes (overhead throwing especially), inflammatory conditions
  - Older population – overuse/degeneration very common

# Location, Location, Location

- Anatomy is so key!
  - LH of biceps tendon lives in close proximity to rotator cuff tendons
  - Weaves its way through tight intertubercular groove
  - Susceptible to impingement
  - Originates on the supraglenoid tubercle AND the superior aspect of the glenoid labrum (so labral problems = biceps problems)
  - Proximal tendon is an area of relative hypovascularity






# Clinical Presentation

- History
  - Acute (tear or inflammation) or chronic (degenerative, overuse) onset
  - Anterior shoulder pain
- Physical
  - Tender to palpate (feel tendon in intertubercular groove)
  - Worse with shoulder flexion, elbow flexion (particularly with forearm in a supinated position)
  - Weakness if frank tear
    - Visible defect due to muscle retraction





# Diagnosis

- Clinical examination often most helpful
- Differential diagnosis
  - RC tendinopathy
  - Subacromial bursitis
  - Labral tear ("SLAP lesion")
  - Shoulder or AC joint arthritis
  - Cervical radiculopathy
  - Referred pain
    - Facetogenic?
    - Visceral presentation with shoulder pain?
- Investigations
  - US helpful if unclear
    - Allows simultaneous assessment of RC tendons and subacromial bursa
  - MR Arthrogram if suspicious of SLAP lesion
    - ("superolateral anterior to posterior" tear of the labrum)

# Management

- Education
  - Prognosis depends on nature of injury
    - Vast majority are chronic, and symptomatically improve with conservative management
- Non-pharm
  - Physiotherapy is mainstay of treatment
    - May use modalities for acute pain but active therapy is necessary!
  - Relative rest (overhead activity)
- Pharm
  - Analgesia as needed
- Intervention
  - Surgical options can be considered for refractory cases